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Research Article

Cellular automaton modeling of austenite formation from ferrite plus pearlite microstructures during intercritical annealing of a C-Mn steel

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A mesoscopic cellular automaton model was developed to study the microstructure evolution and solute redistribution of austenization during intercritical annealing of a C-Mn steel. This model enables a depiction of three-stage kinetics of the transformation combined with the thermodynamic analysis: (1) the rapid austenite growth accompanied with pearlite degeneration until the pearlite dissolves completely; (2) the slower austenite growth into ferrite with a rate limiting factor of carbon diffusion in austenite; and (3) the slow austenite growth in control of the manganese diffusion until the final equilibrium reached for ferrite and austenite. The effect of the annealing temperature on the transformation kinetics and solute partition is also quantitatively rationalized using this model.

Keywords:

Austenization; Intercritical annealing; C-Mn steels; Cellular automaton; Mesoscale modeling.

1. Introduction

Austenization during the intercritical annealing is a research hotspot in automotive manufacturing for its profound value to advance the mechanical properties of the final dual-phase (DP) steels product [1]. During past decades, substantial amount of work has been performed to study the intercritical austenite formation mechanism [2-4], with particular emphasis lay on the DP steels [5]. The austenite formation from ferrite-pearlite aggregates usually includes two transformation stages, i.e. the fast pearlite dissolution and the subsequent slow austenite formation from ferrite. In the first step, the newly formed austenite rapidly overgrows the prior pearlitic regions which is under

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control of the carbon diffusion; whereas the subsequent austenite to ferrite transformation is much slower and under control of carbon diffusion in austenite and the slow diffusion of substitutional alloying elements [6]. The partitioning of substitutional alloying elements within the mobile interface might be accompanied during the subsequent annealing process for its influential role in kinetics of the austenization from ferrite [7, 8].

With the recent development of mesoscopic transformation model, microstructure modeling now can greatly aid advancement of knowledge on the phase transformation mechanism as well as the morphological complexity [9-14]. However, modeling of the austenite formation during the intercritical austenitization is much less advanced compared to the austenite decomposition. There are a few microstructure-based simulation models, e.g. the cellular automaton models (CA) [10, 15-19], phase field models [11] and the Monte Carlo models [20] are available for austenitization simulation. Yet, the full austenitization upon the reheating process that previous researcher chosen were not necessarily appropriate for the intercritical annealing practice. Jacot et al. established a eutectoid lamellar pearlite dissolution model in Fe-C binary system considering curvature effect [15]. Vijay et al. made an assumption that austenite growth is only controlled by carbon diffusion during the heating process [18]. In addition, from a thermodynamic point of view, occurrence of introduced substitutional atom partition between ferrite and newly formed austenite would extremely complicate the model when expanding the model to ternary system [21]. Bos et al. proposed a three-dimensional CA model which is suitable for the simulation of partial austenite formation during the entire intercritical annealing process in DP steels [19, 22]. However, in their modeling, the investigated DP steel is treated as a pseudo-binary alloy. Their model thereof cannot describe the partitioning transformation during the austenite growth into ferrite.

On account of different hypotheses for substitutional elements partitioning mode, there have been two main models to depict the interfacial thermodynamic conditions in ternary Fe-C-*X* alloys, where *X* represents the substitutional alloy element (*X*=Mn, Mo, Cr, Ni, etc.): the first relies on the constrained equilibrium, known as *para*-equilibrium (PE) [23]. Under this condition, it is assumed that no redistribution of the substitutional alloying elements *X* is required during the phase transformation process. The kinetics of the phase transformation is only related to the carbon diffusion. The second is based on the local equilibrium (LE) concept [24], in which both the interstitial and substitutional atoms are supposed to be in full local equilibrium and redistribute within the interface. If the driving force is large enough, the interface would migrate without bulk *X* redistribution. The phase transformation would proceed in a high speed and under control of carbon diffusion. But local chemical equilibrium for C and *X* can still be maintained in the vicinity of the moving interface. This mode is termed as local equilibrium with negligible partitioning (LENP) [23]. When the driving force is small, the phase transformation must be proceeding together with bulk *X* redistribution. The transformation kinetics would be very slow, being effectively under control of the alloying elements diffusion. This mode is referred to as local equilibrium with partitioning (LEP) [23]. In the last decade, these two models have got extensive use in the field of the growth kinetics description for partitioning phase transformation in Fe-C-*X* alloys. Rudnizki et al. used commercial phase field calculation software *MICRESS* to simulate austenite formation in PE and LENP mode, in which the partitioning of slow diffusive substitutional elements is implemented by modified order parameter [25].

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In this work, a modified cellular automaton (CA) model was proposed to describe various transformation stages of the microstructure evolution and solute redistributions (C and Mn) during the intercritical austenization of a Fe-C-Mn alloy. Close attention was paid on the influence of substitutional alloying elements on the transformation kinetics of austenization. By combining with the thermodynamic theories, this CA model makes kinetics switching from fast austenite growth without Mn partitioning to sluggish austenite growth with Mn partitioning to bring off. The influence of the annealing temperature on the transformation kinetics and microstructure development is also rationalized by this model.

2. Model formulation

2.1 Austenite formation

Previous studies on mechanism and kinetics of the austenitization have suggested that the intercritical austenitization transformation consists of several distinct scenarios, i.e. the rapid dissolution of the pearlite, the fast austenite growth into ferrite unaccompanied by the Mn redistribution and the very slow austenite growth with Mn partition [26]. In order to precisely describe the phase transformation in the entire procedures, a mixed-mode concept, which is previously presented for binary Fe-C alloy by Sietsma et al. [27, 28] and further extended to Fe-C-*X* ternary alloy with partitioning elements (*X*=Mn, Cr, Ni, Si, etc.) [29], is introduced to simulate the austenite growth in present model. Both the interface mobility and the alloying element diffusivity are regarded in limited quantities based on the mixed-mode model. In general, the interface velocity of a sharp interface model can be formulated as follows [28]:

$$v_{\alpha\gamma} = M_{\alpha\gamma}\Delta G \qquad (1)$$

where $M_{\alpha\gamma}$ represents the finite interface mobility, and ΔG is the chemical driving force term. The mobility is supposed to obey an Arhenius relationship as follows [19]:

$$M_{\alpha\gamma} = M_0 \exp(-Q_{\alpha\gamma}/RT) \qquad (2)$$

where $Q_{\alpha\gamma}$ is the activation energy with the magnitude of 140 kJ mol^{-1}; M_0 is the mobility pre-exponential factor with a value of 0.5 m mol J^{-1}s^{-1}. R is the universal gas constant (8.314 J mol^{-1} K^{-1}). T is the absolute temperature. For the α→γ phase transformation process, ΔG can be formulated by Eq. (3) [29]

$$\Delta G = \sum_{i=1}^{K} x_i^{\gamma} \cdot (\mu_i^{\alpha} - \mu_i^{\gamma}) \qquad (3)$$

where K is the number of alloying elements in the system. x_i^{γ} represents the concentration of component i (i = Fe, Mn, etc.) in the forming austenite phase. $\mu_i^{\gamma}, \mu_i^{\alpha}$ are the chemical potentials of the component i in separate sides of phase boundary. They are a function of temperature and local interface concentration on austenite and ferrite side of the phase interface. For the sharp interface assumption that boundary width is negligibly thin compared to grain size, it is reasonable to expect a continuous chemical potential of the fast diffusion solute across the phase boundary [30]. Therefore, similar with the treatment by Bohemen et al. in the mixed-mode model [31], the chemical potential difference of interstitial atom carbon across the interface is neglected in present model. The expression of driving force is only relevant to the

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substitutional elements, Fe and Mn.

The substitutional element Mn is supposed not to redistribute among ferrite and austenite within the phase interface at the early stage of the austenite formation [26]. In order to describe ferrite degeneration at the same time, the constrained equilibrium of *para*-equlibrium (PE) theory is assumed as the governing mechanism. The system kinetics is determined by the carbon diffusion, which hence produces relatively high transformation velocity. Under such circumstance, the chemical driving force can be calculated by Eq. (4):

$$\Delta G = x_{\mathrm{Fe}}^{\gamma}(\mu_{\mathrm{Fe}}^{\alpha} - \mu_{\mathrm{Fe}}^{\gamma}) + x_{\mathrm{Mn}}^{0}(\mu_{\mathrm{Mn}}^{\alpha} - \mu_{\mathrm{Mn}}^{\gamma}) \quad (4)$$

where, $\mu_{\mathrm{Fe}}^{\gamma}$, $\mu_{\mathrm{Fe}}^{\alpha}$ are the chemical potentials of iron atom at either side of γ/α phase boundary, respectively. They are decided by the carbon concentration level on the austenite and ferrite interface based on the mass conservation and can be evaluated using standard thermodynamic models. Here, the regular solute sub-lattice model for calculating the chemical potentials in different phases is used [32]. $\mu_{\mathrm{Mn}}^{\gamma}$, $\mu_{\mathrm{Mn}}^{\alpha}$ represent the chemical potentials of manganese atom in austenite and ferrite at the initial Mn concentration x_{Mn}^{0}. They are determined from a *Thermo-Calc* calculation [33], as shown in **Fig. 1**.

In this simulation, the pearlite with relatively small dimension is disposed as a single phase. In order to describe the rapid transformation kinetics of pearlite dissolution into austenite, the driving force at the maximum attained carbon concentration (eutectoid concentration) is chosen, which ensures that pearlite transforms into austenite quite faster while exhibiting similar temperature dependence.

With the carbon redistribution in austenite, the driving force for the transformation is reduced dramatically. When the interface velocity decreases to an extent that the slow Mn diffusion can keep pace with the interface migration, the Mn will be partitioned between α and γ phases. The moment for triggering the partition of Mn is determined by considering a comparison of the time needed for Mn to diffuse a distance with the time required for the interface to travel the same distance [34]. When the velocity of the interface is smaller than the diffusion speed of Mn, Mn atoms would transfer from α into γ across the interface driven by the chemical potential difference at different phase, as seen in **Fig. 1**. The transformation kinetics could be intrinsically determined by the Mn diffusion, leading to a slower reaction velocity. Here, the model that proposed by Hutchinson et al. is adopted to describe the state transition from PE to LEP and the associated change of interfacial solute concentrations. Under this circumstance, the driving force for interface moving is postulated to be as follows [34]:

$$\Delta G = \frac{(x_{\mathrm{Mn}}^{\alpha} + x_{\mathrm{Mn}}^{\gamma})}{2} \cdot (\mu_{\mathrm{Mn}}^{\alpha} - \mu_{\mathrm{Mn}}^{\gamma}) + \frac{(x_{\mathrm{Fe}}^{\alpha} + x_{\mathrm{Fe}}^{\gamma})}{2} \cdot (\mu_{\mathrm{Fe}}^{\alpha} - \mu_{\mathrm{Fe}}^{\gamma}) \quad (5)$$

Therefore, a flux of Mn atoms that is attributed to the chemical potential difference across the phase interface is described by Eq. (6) [34]:

$$J_{\mathrm{Mn}}^{\mathrm{Int}} = \frac{x_{\mathrm{Mn}} \cdot M_{\mathrm{Mn}}^{\mathrm{Trans-int}}}{V_{\mathrm{m}}} \frac{(\mu_{\mathrm{Mn}}^{\alpha} - \mu_{\mathrm{Mn}}^{\gamma})}{\delta} \quad (6)$$

where $M_{\mathrm{Mn}}^{\mathrm{Trans-int}}$ represents the Mn mobility across the interface, V_{m} is the molar volume, and δ denotes the limited interface boundary thickness. It is worth to note that the interface thickness is only implemented for the $J_{\mathrm{Mn}}^{\mathrm{Int}}$ calculation. In a usual sharp-interface CA model, the interface thickness is neglected. With the interface moving, the carbon concentration will be redistributed concurrently for mass balance, which results

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in a carbon flux $J_{\mathrm{C}}^{\mathrm{Int}}$across the interface

$$J_{\mathrm{C}}^{\mathrm{Int}} = (x_{\mathrm{C}}^{\gamma} - x_{\mathrm{C}}^{\alpha}) \cdot v_{\alpha\gamma} \quad (7)$$

The new austenite grains are supposed to nucleate within the original pearlite regions with the bulk Mn concentration. A time-depended continuous nucleation law is implemented to describe the austenite nucleation, which is similar to that used in Ref. [16]. The nucleation probability is assumed to be uniform for each pearlite cell. Details of the nucleation model can be found in Ref. [35].

2.2 Solute diffusion

With a mixed-mode assumption, the austenite formation related solute diffusion and the interface dynamics can both be depicted as a free-boundary problem. The solute diffusion in two phases can be described by Eq. (8):

$$\frac{\partial x_i^{\varphi}}{\partial t} = \nabla \cdot (D_i^{\varphi} \nabla x_i^{\varphi}) \quad (8)$$

where φ denotesαorγphase and i represents C or Mn element.x_i^{φ} is the concentration of species i in phaseφ. D_i^{φ} denotes the diffusion coefficient of *C* or Mn inαorγphase. The diffusion is a thermally activated procedure thus coefficient D_i^{φ} can be formulated as follows:

$$D_i^{\varphi} = D_{i,0}^{\varphi} \exp(-Q_i^{\varphi}/RT) \quad (9)$$

with $D_{i,0}^{\varphi}$ the pre-exponential constant and Q_{i}^{φ} the activation energy for diffusion. These key diffusion parameters are listed in Table 1. At 760 °C, for example, the diffusion coefficients are calculated to be D_{Mn}^{γ}= 5.605×10^{-19} m^2 s^{-1}, D_{Mn}^{α}=2.095×10^{-16} m^2 s^{-1} andD_{C}^{γ}=1.665×10^{-12} m^2 s^{-1}. As a result of the orders of magnitude discrepancy in the diffusivities between the substitutional Mn element and C atoms, the local equilibrium at the interface for both C and Mn is hard to satisfy at the same time during the austenite formation. Thus, the phase transformation is expected to progress in various transformation stages with or without Mn partitioning.

2.3 Grain coarsening

Boundary curvature effect derived competitive growth of the austenite grains would cause grain coarsening after the austenite grain fronts impinge upon each other. The velocity of a grain boundary caused by the interface curvature can be described by Eq. (10)[37]

$$v_{\gamma\gamma} = M_{\gamma\gamma} P \quad (10)$$

where $M_{\gamma\gamma}$ represents the grain boundary mobility across the grain interface and P denotes the driving force for interface migration. The grain boundary mobility driving force P is formulated as

$$P = \eta\kappa \quad (11)$$

where η is the interfacial energy of austenite grain and κ is the interfacial curvature of the corresponding interface segment. In the current model, the grain boundary curvature is calculated by a numerical method based on a lattice model [38],

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$$\kappa = \frac{A}{C_{\text{cell}}} \frac{\text{Kink} - N_i}{N+1} \quad (12)$$

where $A = 1.28$ signifies a topological coefficient, C_{cell} denotes the grid spacing of the cellular automaton model, $N = 18$ counts the number of the first and second nearest neighbors for a hexagonal lattice, N_i is the number of cells within the neighborhood belonging to grain i, and $\text{Kink} = 9$ is the number of cells within the neighborhood belonging to grain i under the circumstance of a flat interface ($\kappa = 0$). The topological philosophy behind this model can be found in Figs. A1–A3 in Ref. [39].

3. Cellular automaton model

The Cellular automaton (CA) model used here for the present intercritial austenization calculation is developed from the method proposed in the previous study [40]. In the current CA model, the 2D spatial system is discretized into a number of equal spaced regular hexagonal lattices. Each cell lattice represents a certain part of real material which is featured by several attributes such as distinct phase index, separate solute content and an orientation indicator. The Von Neumann's rule is used here to define the neighbors of a cell, which only takes the nearest six cells into consideration. In 2D CA model, the autonomous variables are spatial position $X=(x, y)$ and time t. In order to specifically describe the intercritical austenization phase transformation procedure, seven characteristic state variables are assigned to every last CA element. They are: (i) the phase state variable that distinguishes whether the cell belongs to pearlite, austenite, ferrite, or is a part of γ/α, P/γ interface; (ii) the carbon concentration fraction; (iii) the manganese concentration state; (iv) the grain orientation variable is randomly allocated integer standing for its crystallographic orientation; (v) the transformed austenite fraction, f_γ, representing the austenite transformation fraction originated from ferrite; (vi) the other austenite transformation fraction, $f_\gamma^{'}$, quantifying the austenite fraction stem from pearlite; (vii) the coarsening fraction variable, f, used to depict the boundary segment migration between the γ/γ grain interface. The state variables on every CA element are a function of previous state of itself and its neighbor cells. They are evolving based on the aforementioned sub-models in Section 2. The automaton kinetics is accomplished by synchronously updating the discrete state variables for all CA cells at every time step.

In this CA model, state transition takes place only in the interface or grain boundary cells. At t moment, the migration distance for a movable cell (i, j) within one-time step, Δt can be depicted as

$$l_{i,j}^{t+\Delta t} = l_{i,j}^{t} + v_{i,j}^{t} \Delta t \quad (13)$$

where, $v_{i,j}^{t}$ represents the velocity of the boundary migration during austenite growth or grain coarsening. The subscripts (i, j) denote the space coordinates of the chosen interface cell. Then the transformation fraction of the selected cell, $f_{i,j}^{t}$ can be calculated by Eq. (14):

$$f_{i,j}^{t} = l_{i,j}^{t} / L_{\text{CA}} \quad (14)$$

where L_{CA} is the grid spacing. The interface cell would transfer into a new state

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determined by the switching rule when the transformation fraction is above 1.0.

Detailed rules about accomplishment of the physical metallurgical basis in the CA model such as nucleation, growth and grain coarsening process during the phase transformation, can be found in Ref.40 and 41. Yet, only the specific relevant parts for the current austenization calculation model are listed as follows:

(1) In this simulation, pearlite is regarded as one effective phase with a constant carbon concentration of 0.7 wt.% on the basis of *Thermo-Calc* database.

(2) The austenite growth from pearlite is supposed to under the control of carbon diffusion only with the driven force derived from the maximum available carbon concentration (0.7 wt.%). This assumption could ensure a much faster transformation of pearlite dissolution than subsequent α to γ transformation.

(3) When the driving force contributed by the difference of the Fe chemical potentials in both phases is lower than that from Mn, the system will start to deviate from PE mode and the Mn solute spike will gradually heap up. This is assumed to be the criterion implemented in current simulation to describe the transition from the non-partitioning transformation to partitioning transformation.

(4) In the early stage of the phase transformation procedure under PE mode, the γ/α interface migrates without Mn partitioning, i.e. $J_{\mathrm{Mn}}^{\mathrm{Int}}$ =0. The carbon flux $J_{\mathrm{C}}^{\mathrm{Int}}$ through the grain boundary is calculated by Eq. (7). When the partitioning transformation is triggered, the flux of Mn across the moving interface $J_{\mathrm{Mn}}^{\mathrm{Int}}$ is calculated according to Eq. (6); and the carbon flux $J_{\mathrm{C}}^{\mathrm{Int}}$ is determined with Eq. (7).

(5) Curvature driven grain coarsening procedure is supposed to occur solely between the austenite grain boundaries after the hard impingement occurs.

4. Simulation setting

The material used in the research is a typical C-Mn steel with a chemical composition of Fe-0.08C-1.75Mn (in wt.%). The Ae3 temperature is calculated as 816 °C with the thermodynamic database *Thermo-Calc*. The steel is usually annealed within the intercritical temperature ranges of Ae1~Ae3 to produce adequate ferrite-austenite compound, subsequently followed by proper cooling to form the final ferrite-martensite dual-phase structure. This study is intended to present numerical modeling of the important austenization phase transformation within the isothermal intercritical annealing procedure.

Construction of the phase transformation simulation requires pre-definitions of the computational domain and the starting microstructure including phase distribution and initial concentrations in each phase. The simulation in this work is carried out on a 350×300 2D lattice with 0.4 μm grid spacing, corresponding to a physical domain of 140 μm × 120 μm in material entity. The starting microstructure used for current modeling is generated from a separate cellular automaton simulation, as shown in **Fig. 2**. The initial microstructure morphology of the model consists of a fine, unbanded ferrite-pearlite aggregate with about 10 pct of pearlite in volume. In the figure, the white regions are the ferrite phases and the dark gray regions represent the pearlite colonies. The black lines indicate the α/α grain interfaces or the α/P boundaries. The original average ferrite grain size is about 15 μm in diameter. In current modeling, pearlite is regarded as one individual phase, as treated in Ref. [18], with a carbon concentration

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of 0.7 wt.%. The initial carbon concentration of ferrite phase is setting as 0.01 wt.%. The carbon concentration of ferrite has less impact on the transformation dynamics owing to its exceedingly low carbon solubility. Hence, carbon diffusion within ferrite phase can be disregarded in this simulation. A unique initial manganese concentration of 1.75 wt.% is given both in the pearlite colonies and in the ferrite matrix. The key parameters used in this calculation model are listed in Table 2.

5. Results and discussion

Fig. 3 displays the calculated austenite formation kinetics at the intercritical annealing temperature of 760 °C. Corresponding microstructure morphology and the solute field of manganese and carbon evolution are shown in Fig. 4. In the simulated microstructure maps, the dark gray areas denote the un-dissolved pearlite, the orange are the newly formed austenite and the white are the ferrite phase. In the carbon concentration profiles, the light gray regions with the lowest carbon solubility indicate the ferrite phase. From Figs. 3 and 4, it can be seen that the phase transformation occurs in several transformation stages, as observed in many experimental findings [6-8, 26].

Austenite formation begins from the original pearlite aggregates by nucleation. Afterwards, newly formed austenite nuclei would grow up quickly with the degeneration of pearlite, as seen in Fig. 4(a). Generally, the austenite growth with costing pearlite is mainly controlled by carbon diffusion within austenite phase because of a very limited diffusion distance approximately equal to the pearlite interlamellar spacing [6]. The growth rate is thus expected to be extremely rapid. However, as mentioned previously, the grid spacing used in this simulation is too large to describe the pearlite lamellae separately. Pearlite is considered as an individual phase in the simulation. The driving force at the maximum attainable carbon concentration is used to describe the growth rate of austenite within the pearlite colonies. This treatment could ensure that pearlite transforms into austenite quite fast while exhibiting similar temperature dependence. As a result, an apparent inflection arises in the transformation kinetics curve between the two transformation stages owing to the difference in the transformation rate. This result is consistent with many experimental findings [6, 8]. The associated solute changing in this transition can be found in Fig. 5 which schematically illustrates various transformation procedures that appear in sequence during the austenite formation phase transformation.

With pearlite dissolution accomplishment, newly formed austenite would be filled with a relatively high carbon concentration, which is not in equilibrium with the ferrite phase in thermodynamic. Consequently, the austenite would continue to grow up outward into the surrounding ferrite with the purpose of achieving equilibrium with the ferrite constitutes at this temperature. This process might be accompanied either with or without manganese partition according to the driving force. At the beginning stage just after pearlite dissolution, the high carbon concentration of austenite introduces high driven force for the γ/α interface movement, as seen in Fig. 4(b). Under such condition, the interface would move quickly accompanying with barely any substitutional element partition due to its extraordinarily low diffusivity compared to carbon. The phase

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transformation hence advances following the *para*-equlibrium mode, i.e. the austenite growth takes place without any redistribution of Mn at the mobile γ/α interface and thus the austenite formation is under the control of the carbon diffusion within austenite. The schematic illustration of this process is shown in Fig. 5(b).

Fig. 6(a) displays the variation of the carbon distribution along with the red straight line (starting from the left) shown in Fig. 4. The carbon concentration at the austenite side interface decreases rapidly approaching the equilibrium with the continuous austenite growth into ferrite. Significant carbon gradients arise within the austenite phase. This would cause a drastic decrease in the driving force and hence the interface velocity at the moving interfaces. Consequently, the transformation kinetics would become slower than that at the first stage of rapid pearlite dissolution as shown in Fig. 3. Meanwhile, homogenization of carbon distribution is accompanying with the phase transformation in austenite driven by the carbon concentration gradient. The carbon gradients are found less and less noticeable with the phase transformation proceeding. At the late stage of 75 s, although the carbon concentration within each austenite grains has homogenized considerably, significant concentration difference still exists between different austenite grains. In addition, extreme carbon concentration gradients are maintained in part of large austenite colonies on account of the finite carbon diffusivity in austenite. These results display the inhomogeneous feature of the distribution of carbon concentration quite informatively, which will have an effect on the transformation kinetics during the subsequent heat-treating procedure and the mechanical properties of the final product.

Fig. 7 shows the isothermal section at 760 ^{o}C with equilibrium tie line calculated by *Thermo-Calc*. The possible austenite solute concentration evolution paths are depicted by the black dotted lines (C_0 is the initial composition of the investigated steel). The austenite growth with the ferrite degradation would proceed towards the PE limit under the control of carbon diffusion at the early stage. However, at the late stage when the interfacial condition nearly arrives at the PE limit, migration of the γ/α interface is slowed severely such that Mn partition becomes possible. Driven by the chemical potential difference derived from Mn, as shown in Fig. 1, Mn atoms would jump across the γ/α interface from ferrite side to the austenite side. A net flux of Mn atoms across the phase interface would come into being and lead to a gradual buildup of Mn spike, as seen in Fig. 6(b). Concomitantly, significant depleted spike of Mn is developed at the ferrite side, which brings about a zigzag Mn-profile on the interface as schematically illustrated by Fig. 5(c). This is the sign that the system is departing from the PE. Afterwards, the interfacial conditions are evolving towards those of local equilibrium by following the concentration path e→d in γ side and e'→d' in α side at the moving interface, as seen in Fig. 7.

With increasing of the Mn spike, the chemical potential difference of Mn atoms across the phase boundary diminishes gradually. The driven force for interface movement decreases accordingly and thereby slows down the phase transformation velocity. This implies a solute retarding effect exerted upon the moving interface correlated with the gradual buildup of the zigzag Mn profile. The growth kinetics

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changes to be controlled by the Mn partitioning at this stage. On the other hand, on account of the Mn partitioning during the interface migration, the phase interface condition has to be adapted so that the mass balance (including Mn and C gradients) is in accordance with the velocity. The duration of the transition period of the interface condition from para-equilibrium to local equilibrium should also be determined by the time took for adjustment of the local conditions at the interface. This will persist for a substantially long period.

When the interfacial Mn concentration in the austenite is increased up to the ortho equilibrium tie-line endpoint determined composition shown as "d" in Fig. 7, the driven force of the interface would fall to zero. The system then reaches the local equilibrium condition with Mn spikes on each side of the interface counterbalanced with each other. Further austenite growth would proceed very slowly under the control of bulk Mn diffusion in both phases until the final equilibrium accomplished as indicated in Fig. 5(d). However, actually, the final bulk equilibrium is hard to reach in practical thermal treatments due to the extremely low Mn diffusivity in austenite. As a result, Mn enrichment rims appear around the austenite patches caused by the Mn partitioning across the γ/α interface. In microstructural characterizations, the Mn partitioning is usually evidenced by existence of the "martensite rings" around other decomposition products of the austenite [43] (e.g. the pearlite). It should be noted that, in practice, the partitioning phase transformation does not contribute to the overall transformation fraction so much significantly. However, the Mn partitioning is very important for quenching of the steel since it can locally increase the hardenability of the austenite [4]. The partitioning transformation highly depends on local interfacial conditions. Due to the microstructure inhomogeneity, the partitioning transformation occurs heterogeneously within the entire simulation domain, as seen in Fig. 4. Mn partitioning in smaller austenite patches is found to start much sooner before the carbon-controlled process is completed within the larger austenite colonies. This result implies that the overall phase transformation should be jointly controlled by concomitance of the two mechanisms at this stage. Therefore, a long transition period from carbon diffusion controlled to Mn diffusion controlled does exist implicitly within the overall kinetics in Fig. 3, which is strongly dependent on the grain size distribution. However, it ought to be pointed out that current work is intended to develop an informative mesoscopic model for the intercritical austenitization, describing the various transformation steps of the microstructural changing. Calculating of the final equilibrium is not in the range of current work due to the long transformation time.

Fig. 8 displays the temporal evolutions of the austenite fraction at various annealing temperatures (T=740, 760 and 780 °C). Rapid pearlite dissolution at the first transformation stage occurs at all three temperatures. Noticeable transitions from rapid to slow transformation rates are also visible in the transformation kinetics when pearlite dissolution is completed. However, the subsequent austenite transformation is promoted significantly at higher annealing temperature. This indicates that increasing the annealing temperature is effective for austenite formation owing to the diffusional transformation mechanism. It is known that the austenite carbon solubility deceases

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with the temperature increasing. Thus, the carbon concentration needed for austenization is much lower at a higher annealing temperature. It makes the phase transformation easier and consequently leads to a higher austenite fraction.

Fig. 9 depicts the resultant microstructures and the corresponding solute fields both of manganese and carbon when t =75 s at the three intercritical temperatures. It can be observed that a higher austenitic fraction is produced with a higher annealing temperature. And a lower carbon solubility in austenite leads to larger austenite colonies. A closer look at the enlarged Mn concentration fields in Fig. 9 of the white quadrangles at the three temperatures indicates that Mn partitioning takes place more severely at the lower annealing temperature. The predicted tendency is in accordance with the results of Speich et al. [8], who found that the phase transformation tends to under the control of the Mn diffusion at low temperature, while affected by the carbon diffusion at high temperature. Their results could be understood as the consequence of the temperature-depended transition from non-partitioning transformation to partitioning transformation suggested by present simulation.

6. Conclusions

In this work, a cellular automaton modeling has been performed to simulate the austenization transformation during the intercritical annealing of a C-Mn steel. By combination with thermodynamically based analyses, this model enables the depictions of the Mn-partitioning austenite formation as well as phase transformation mode switching from fast non-partitioning austenite growth towards sluggish Mn-partitioning growth. The main conclusions are:

(1) Beginning with a ferrite plus pearlite initial microstructure, various transformation stages of the intercritical austenite formation can be depicted, i.e. (i) rapid austenite growth with pearlite costing until pearlite dissolution completion; (ii) slower austenite growth with ferrite degradation under the control of carbon diffusion within austenite; and (iii) much sluggish growth of austenite subjected to manganese diffusion and the final equilibration achievement.

(2) The transition from non-partitioning transformation to partitioning transformation occurs in a quite broad range of time frames within the entire microstructure, highly affected by the inhomogeneity of the microstructure. This will accordingly influence overall transformation kinetics of the intercritical austenite formation.

(3) Increasing the annealing temperature is conducive to the austenite formation owing to the diffusional transformation nature. However, the Mn partitioning process might be delayed, depending on the driving force.

Acknowledgement

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Chinese Academy of Sciences (No. 2016176).

References

1. C.C. Tasan, M. Diehl, D. Yan, M. Bechtold, F. Roters, L. Schemmann, C. Zheng, N. Peranio, D. Ponge, M. Koyama, K. Tsuzaki, D. Raabe, Annu. Rev. Mater. Res. 45 (2015) 391-431.
2. H. Azizi-Alizamini, M. Militzer, W.J. Poole, Metall. Mater. Trans. A 42 (2011) 1544-1557.
3. A. Eggbauer, M. Lukas, G. Ressel, P. Prevedel, F. Mendez-Martin, J. Keckes, A. Stark, R. Ebner, J. Mater. Sci. 54 (2019) 9197-9212.
4. M. Enomoto, S. Li, Z.N. Yang, C. Zhang, Z.G. Yang, Calphad 61 (2018) 116-125.
5. V. Singh, M. Adhikary, T. Venugopalan, A. Chakraborty, T. Nanda, B.R. Kumar, Mater. Manuf. Process. 32 (2017) 1806-1816.
6. W.C. Jeong, C.H. Kim, J. Mater. Sci. 20 (1985) 4392-4398.
7. D.Z. Yang, E.L. Brown, D.K. Matlock, G. Krauss, Metall. Trans. A 16 (1985) 1523-1526.
8. G.R. Speich, V.A. Demarest, R.L. Miller, Metall. Mater. Trans. A 12 (1981) 1419-1428.
9. Y.J. Lan, D.Z. Li, C.J. Huang,Y Y Li, Modell. Simul. Mater. Sci. Eng. 12 (2004) 719-729.
10. D. An, S.I. Baik, S. Pan, M. Zhu, D. Isheim, B.W. Krakauer, D.N. Seidman, Metall. Mater. Trans. A 50 (2019) 436-450.
11. B. Zhu, M. Militzer, Metall. Mater. Trans. A 46 (2015) 1073-1084.
12. H. Chen, S. van der Zwaag, Acta Mater. 72 (2014) 1-12.
13. D. An, S. Chen, D. Sun, S. Pan, B.W. Krakauer, M. Zhu, Comput. Mater. Sci. 166 (2019) 210-220.
14. T. Jia, M. Militzer, Metall. Mater. Trans. A 46 (2015) 614-621.
15. A. Jacot, M. Rappaz, R.C. Reed, Acta Mater. 46 (1998) 3949-3962.
16. B.J. Yang, A. Hattiangadi, W.Z. Li, G.F. Zhou, T.E. McGreevy, Mater. Sci. Eng. A 527 (2010) 2978-2984.
17. B.J. Yang, L. Chuzhoy and M.L. Johnson, Comput. Mater. Sci. 41 (2007) 186-194.
18. K. Vijay Reddy, C. Halder, S. Pal, Trans. Ind. Inst. Met. 70 (2017) 909-915.
19. C. Bos, M.G. Mecozzi, J. Sietsma, Comput. Mater. Sci. 48 (2010) 692-699.
20. A. Jacot, M. Rappaz, Acta Mater. 47 (1999) 1645-1651.
21. M. Ollat, M. Militzer, V. Massardier, D. Fabregue, E. Buscarlet, F. Keovilay, M. Perez, Comput. Mater. Sci. 149 (2018) 282-290.
22. C. Bos, M.G. Mecozzi, D.N. Hanlon, M.P. Aarnts, J. Sietsma, Metall. Mater. Trans. A 42 (2011) 3602-3610.
23. H. Chen, C.Y. Zhang, J.N. Zhu, Z.N. Yang, R. Ding, C. Zhang Z.G. Yang, Acta Metall. Sin. 54 (2018) 217-227.
24. G.P. Krielaart, J. Sietsma, S. van der Zwaag, Mater. Sci. Eng. A 237 (1997) 216-223.
25. J. Rudnizki, B. Böttger, U. Prahl, W. Bleck, Metall. Mater. Trans. A 42 (2011) 2516-2525.
26. E. Navara, R. Harrysson, Scr. Mater. 18 (1984) 605-610.
27. M.G. Mecozzi, C. Bos, J. Sietsma, Acta Mater. 88 (2015) 302-313.
28. J. Sietsma, S. van der Zwaag, Acta Mater. 52 (2004) 4143-4152.
29. C. Bos, J. Sietsma, Acta Mater. 57 (2009) 136-144.
30. J. Svoboda, F.D. Fischer, P. Fratzl, E. Gamsjäger, N.K. Simha, Acta Mater. 49 (2001) 1249-1259.
31. S.M.C. Van Bohemen, C. Bos, J. Sietsma, Metall. Trans. A 21 (1990) 2115-2123.
33. J.O. Andersson, T. Helander, L. Höglund, P. Shi, B. Sundman, Calphad 26 (2002) 273-312.

Journal Pre-proof

34. C.R. Hutchinson, A. Fuchsmann, Y. Brechet, Metall. Mater. Trans. A 35 (2004) 1211-1221.
35. M. Umemoto, Z. Hai Guo, I. Tamura, Mater. Sci. Technol. 3 (1987) 249-255.
36. H. Bhadeshia, R. Honeycombe, Steels - Microstructure and Properties, third ed. Elsevier, 2006.
37. C. Zheng, M.N. Xiao, Z.D. Li, Y.Y. Li, Comput. Mater. Sci. 45 (2009) 568-575.
38. R. Sasikumar, R. Sreenivasan, Acta Metall. 42 (1994) 2381-2386.
39. K. Kremeyer, J. Comput. Phys. 142 (1998) 243-263.
40. C.W. Zheng, D. Raabe, D.Z. Li, Acta Mater. 60 (2012) 4768-4779.
41. C.W. Zheng, D. Raabe, Acta Mater. 61 (2013) 5504-5517.
42. C.W. Zheng, N.M. Xiao, L.H. Hao, D.Z. Li, Y.Y. Li, Acta Mater. 57 (2009) 2956-2968.
43. M.M. Souza, J.R.C. Guimarães, K.K. Chawla, Metall. Trans. A 13 (1982) 575-579.

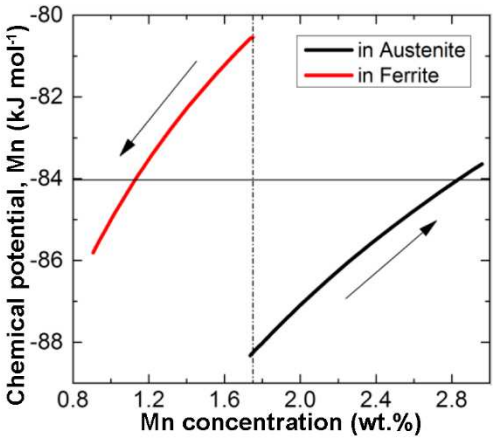

Fig. 1. Chemical potentials of Mn in austenite and ferrite as a function of Mn concentration at 760 °C calculated by Thermo-Calc.

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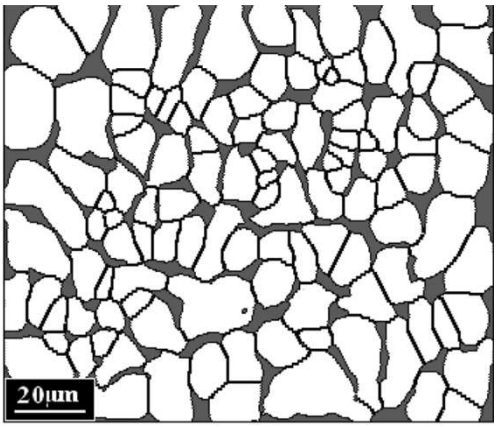

Fig. 2. Initial microstructure for cellular automaton simulation. (White regions: ferrite phase, dark gray regions: pearlite, black lines: grain boundaries.)

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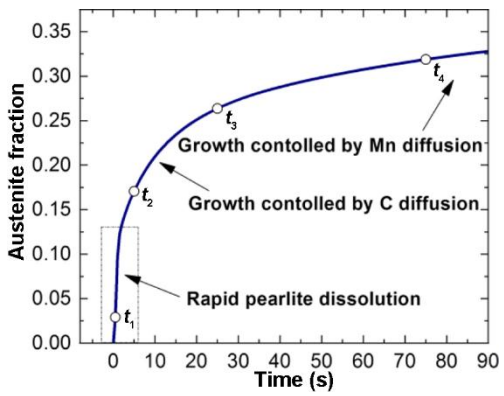

Fig. 3. Simulated kinetics of austenite formation at annealing temperature of 760 ºC.

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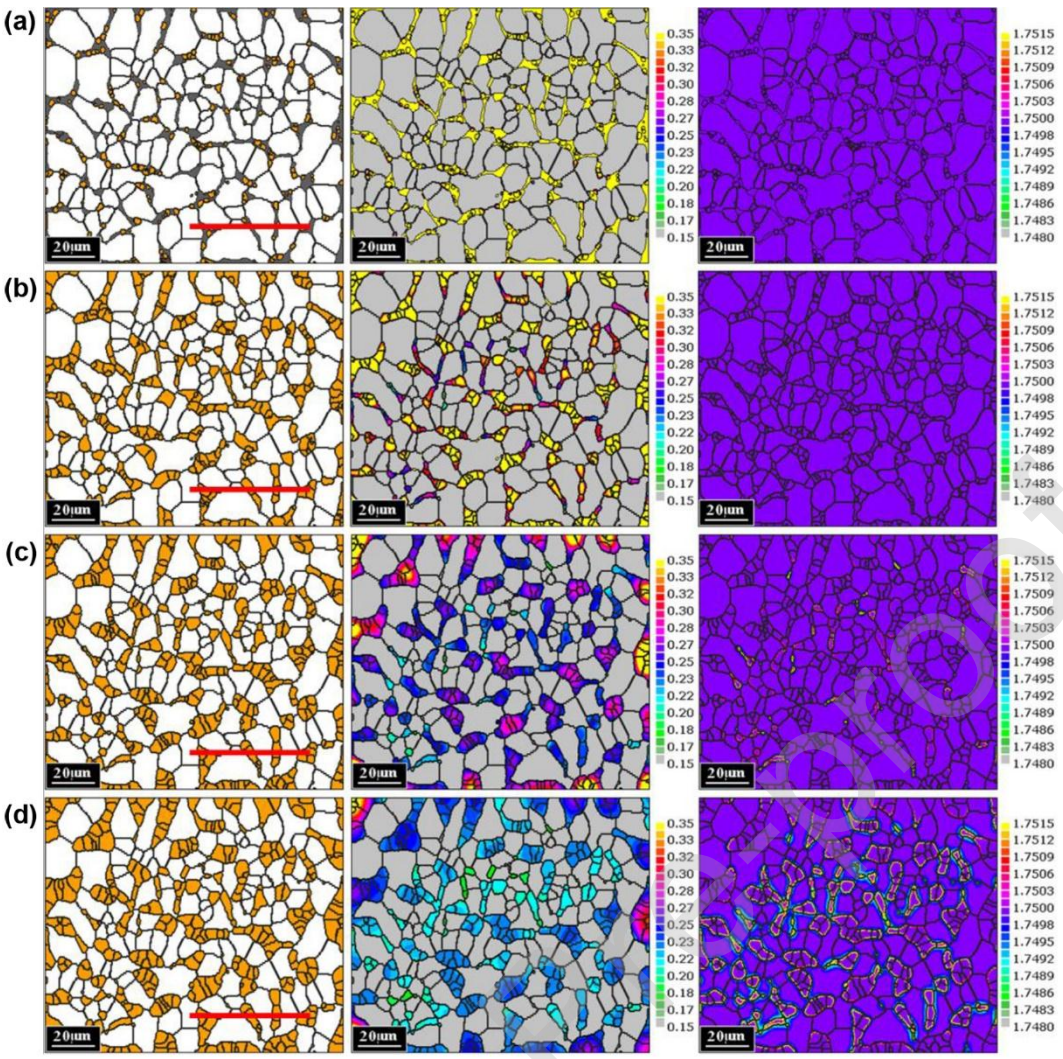

Fig. 4. Simulation results of microstructure evolution (left), carbon concentration distribution (middle) and manganese concentration distribution (right) at the annealing temperature of 760 °C in a Fe-0.08C-1.75Mn (in wt.%) steel: (a) t_1=0.5 s, (b) t_2=5 s, (c) t_3=25 s, (d) t_4=75 s. In the microstructures, the yellow areas are the newly formed austenite; the white regions are ferrite phase and the dark grey regions are pearlite phase. The black lines indicate the grain boundaries.

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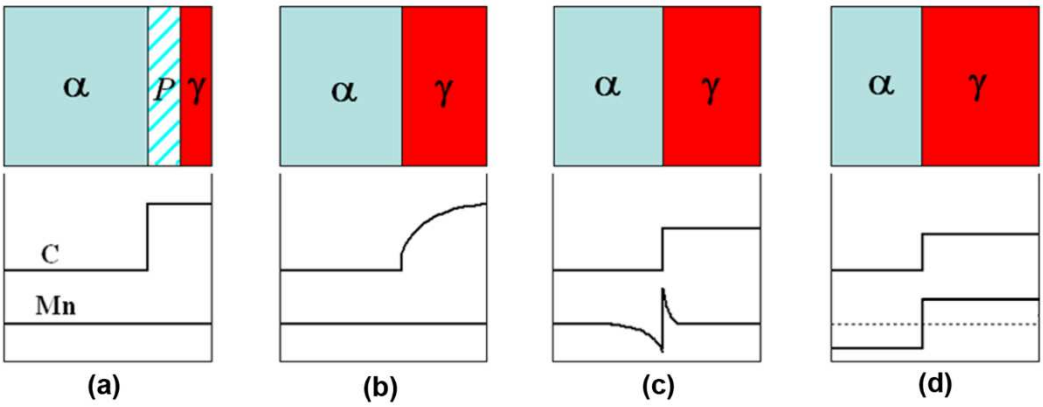

Fig. 5. Schematic diagrams of various transformation stages during austenite growth in intercritical annealing of the C-Mn steel [8]: (a) dissolution of pearlite; (b) austenite growth with carbon diffusion in austenite; (c) austenite growth with manganese partition at the moving interface; and (d) final solute equilibrium.

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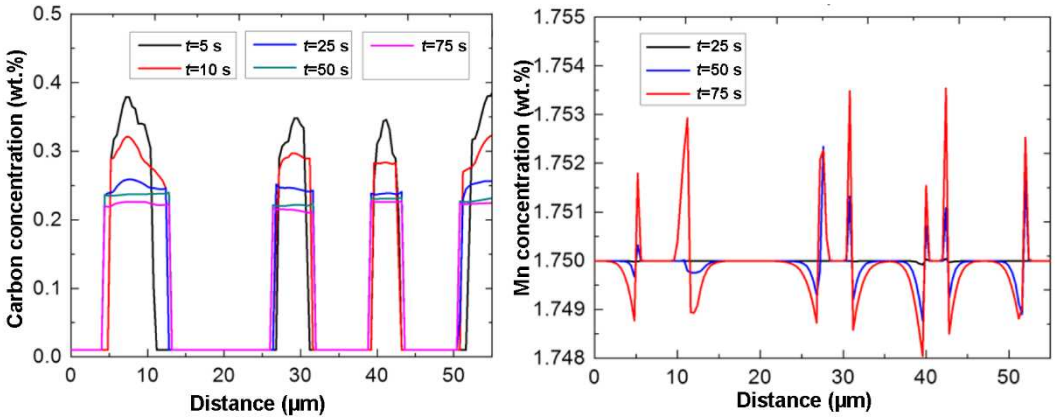

Fig. 6. Evolution of carbon concentration (a) and manganese concentration (b) along the red straight line (starting from the left) shown in Fig. 4.

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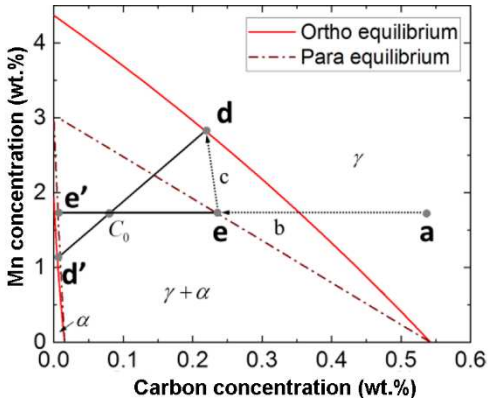

Fig. 7. Isothermal section of Fe-Mn-C ternary system at 760 ºC calculated by Thermo-Calc. The potential concentration path for various transformation stages is indicated with the short-dotted lines.

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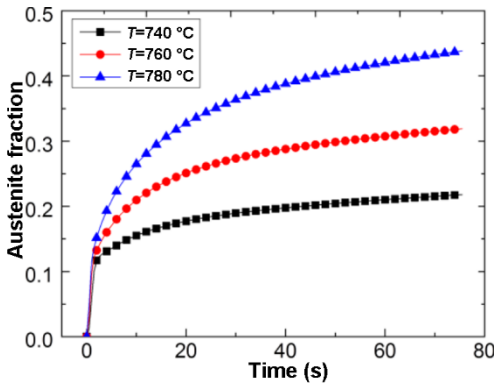

Fig. 8. Simulated transformation kinetics of austenite formation at various annealing temperatures in a Fe-0.08C-1.75Mn (in wt.%) steel.

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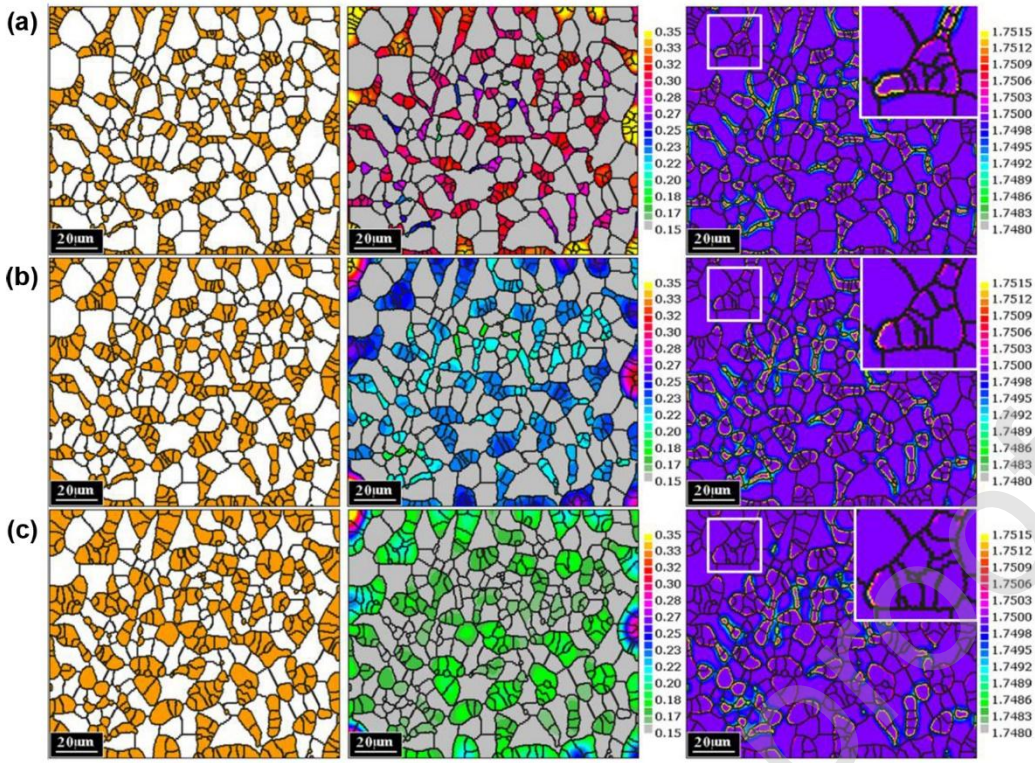

Fig. 9. Simulation results of microstructure (left), carbon concentration field (middle) and manganese concentration field (right) when t=75 s at different annealing temperatures in a Fe-0.08C-1.75Mn (in wt.%) steel: (a) T=740 °C; (b) T=760 °C; (c) T=780 °C. In the microstructures, the yellow areas are the newly formed austenite; the white regions are ferrite phase. The black lines indicate the grain boundaries.

Table 1 Diffusion parameters used in simulation [27, 36].

i	$D_{i,0}^{\gamma}$ (m^2 s^{-1})	Q_i^{γ} (kJ mol^{-1})	$D_{i,0}^{\alpha}$ (m^2 s^{-1})	Q_i^{α} (kJ mol^{-1})
C	2.0×10^{-5}	140	/	/
Mn	5.7×10^{-5}	277	4.36×10^{-4}	243.6

Table 2 Key parameters used in simulation [42].

M_0 (mmol J^{-1} s^{-1})	$Q_{\alpha\gamma}$ (kJ mol^{-1})	$M_{\mathrm{Mn}}^{\mathrm{Trans-int}}$ (m^2 J^{-1} s^{-1})	δ (m)	V_{m} (m^3 mol^{-1})	η (J m^{-2})
0.5	140	2.9×10^{-22}	1×10^{-9}	7.39×10^{-6}	0.56